Keshav Aggarwal

Personal Details

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Education

Jul 2023 – May 2026	Ph.D. Astronomy – **DAASE, Indian Institute of Technology (IIT) Indore** (thesis submitted May 2026) Funding agency: Prime Minister's Research Fellowship Collaborating agency: Space Physics Laboratory, Vikram Sarabhai Space Centre, Thiruvananthapuram Supervisor: Prof. Abhirup Datta *Exploring space weather and its effect on planetary atmospheres using radio occultation*
Aug 2021 – Apr 2023	M.Sc. Astronomy – **DAASE, Indian Institute of Technology (IIT) Indore** Supervisor: Prof. Abhirup Datta *Study of the solar corona using the Indian Mars Orbiter Mission*
Jul 2018 – Jun 2021	B.Sc. Physics, Chemistry and Mathematics – **Chaudhary Charan Singh University, Meerut**

Research Positions

Oct 2023 – Present	**Ph.D. Student Visitor**, Space Physics Laboratory, Vikram Sarabhai Space Centre (ISRO), Thiruvananthapuram
Jul 2023 – May 2026	**Prime Minister's Research Fellow**, DAASE, IIT Indore

Workshops, Schools, and Presentations

Mar 2026	*"A turbulence index independent framework for deriving solar wind speed and coronal electron density from radio spectral broadening"* at **7th Indian Planetary Science Conference (IPSC 2026), IIT Indore, India** (Talk)
Feb 2026	*"Estimating solar wind velocities using spacecraft radio signals"* at **National Space Science Symposium (NSSS 2026), Umiam, Shillong, India** (Talk)
Jan 2026	*"Estimating solar wind speeds using Doppler broadening: results from MOM and Akatsuki"* at **ISRO–ESA Workshop, Thiruvananthapuram, India** (Poster)

Dec 2025	*"Estimating solar wind speeds using Doppler broadening: results from MOM and Akatsuki"* at **AGU Fall Meeting 2025, New Orleans, United States** (Poster)
Dec 2025	*"Estimating solar wind speeds using Doppler broadening: results from MOM and Akatsuki"* at **The Variable Sun, Thiruvananthapuram, India** (Poster)
Sep 2025	*"Probing the Smiley-Sun using Akatsuki radio signals"* at **ASI Symposium 003, Astronomical Society of India, JECRC University, Jaipur, India** (Poster)
Aug 2025	*"Studying the 'Smiley Sun' using the Akatsuki radio occultation experiment"* at **URSI Asia-Pacific Radio Science Conference (URSI AP-RASC 2025), International Convention Centre, Sydney, Australia** (Talk)
Mar 2025	*"On the estimation of solar wind velocity using spectral characteristics of the probing radio signals: results from solar occultation measurements using Indian Mars Orbiter Mission"* at **6th Indian Planetary Science Conference (IPSC), IIT Roorkee, India** (Poster)
Oct 2024	*"Results from radio occultation studies using Indian Mars Orbiter Mission"* at **6th URSI Regional Conference on Radio Science (URSI-RCRS), Graphic Era Hill University, Bhimtal, India** (Poster)
Oct 2024	*"Results from radio occultation studies using Indian Mars Orbiter Mission"* at **3rd Indian Space Weather Conference, IIT Roorkee, India** (Talk)
Oct 2023	*"Unveiling space weather and planetary atmosphere dynamics through Aditya-L1 and DISHA data integration"* at **2nd Indian Space Weather Conference, Physical Research Laboratory, Ahmedabad, India** (Talk)
Sep 2023	*"Retrieving sulphuric acid profiles of the Venus atmosphere from Akatsuki radio occultation data"* at **Venus Science Conference, Physical Research Laboratory, Ahmedabad, India** (Talk)
Aug 2023	**Strange New Worlds conference**, Indian Institute of Science Education and Research (IISER), Pune
Feb 2023	**ISRO–ARIES Aditya-L1 Support Cell Workshop 3**, Indian Institute of Technology (BHU), Varanasi
Jul 2022	**IIA Summer School 2022**, Indian Institute of Astrophysics (IIA), Bengaluru
Jun 2022	**Introductory Summer School in Astronomy and Astrophysics 2022**, Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune
Mar 2022	**Spring 2022: ERS Post-Launch Data Challenge for JWST**, Space Telescope Science Institute, Baltimore, and Heidelberg
Sep 2021	*"HR diagrams of star clusters"* at **Uncover Possibilities programme, Space Advancement and Research Cell, India** (Talk)
Aug 2021	*"The Gaia telescope and the use of its data"* at **Uncover Possibilities programme, Space Advancement and Research Cell, India** (Talk)

Feb 2021	*"Life in the Solar System, and exoplanets"* at **Nakshatra, Astronomy Club, IIT Indore, Indore, India** (Talk)

Achievements

Dec 2025	Recipient of the *AGU Student Travel Grant* for presenting a poster at AGU Fall Meeting 2025, New Orleans, USA
Mar 2025	*Early Career Researcher Award for best poster* at the 6th Indian Planetary Science Conference, IIT Roorkee
Oct 2023	*Prime Minister's Research Fellowship*, Ministry of Education, Government of India

Internships

Aug 2022 – Sep 2023	**Researcher in AI and Orbital Mechanics | HR Coordinator** at ERETS Space London, London (remote) Developed AI/ML and orbital mechanics algorithms for satellites. Handled hiring for the team.
Feb 2022 – Aug 2022	**Data Analyst | Project Coordinator** at Armstrong Space Australia, Melbourne (remote) Built pipelines for acquiring and processing space debris data. Space debris removal by laser ablation. Catalogues of debris between 1 and 10 cm.

Outreach, Service and Miscellaneous

Sep 2024	Volunteer, 8th Aditya-L1 Support Cell workshop, IIT Indore
Aug 2024	Volunteer, National Space Day celebrations organised by DAASE, IIT Indore
Mar 2024	Volunteer, PMRF Symposium, IIT Indore
Feb 2024	Contributed to the Venus mission white paper *'Venus: the perfect laboratory for understanding interactions between space weather and planetary atmospheres'*, submitted to ISRO
Oct 2023	Organised astronomy outreach activities for *100 Hours of Astronomy* (IAU) at IIT Indore
Feb 2023	Organised astronomy outreach activities at *Fluxus*, the IIT Indore annual festival

Technical Skills

Languages	Python (NumPy, SciPy, Astropy, SunPy, Matplotlib), Jupyter
Tools	NASA SPICE toolkit, Git, JWST data reduction pipelines
Data	Planetary radio science data from ISRO, JAXA and ESA missions

Publication List

1. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Banerjee, S., Manikantan, R., Sharma, A., Imamura, T., *A study on the contribution of the interplanetary medium in radio occultation experiments*, **Icarus**, 2026, 10.1016/j.icarus.2026.117291.
2. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Sharma, A., *Plasma turbulence in the lunar environment across solar wind and magnetotail conditions: Observations from Chandrayaan-2 Radio Science experiment*, **Journal of Geophysical Research: Planets**, 2026, 10.1029/2026JE009957.
3. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Banerjee, S., Imamura, T., Ando, H., *A turbulence index independent framework for deriving solar wind speed and coronal electron density from radio spectral broadening*, **Monthly Notices of the Royal Astronomical Society**, 2026, 10.1093/mnras/stag554.
4. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Roopa, M. V., Imamura, T., *A generalized method for estimating solar wind speeds and densities using spectral broadening for a Kolmogorov turbulence spectrum*, **Advances in Space Research**, 2026, 10.1016/j.asr.2026.03.059.
5. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Imamura, T., *On the estimation of solar wind velocity under varying solar activity conditions using Akatsuki measurements*, **Monthly Notices of the Royal Astronomical Society**, 2025, 10.1093/mnras/staf1305.
6. **Aggarwal, K.**, Choudhary, R. K., Datta, A., Roopa, M. V., Dai, B. K., *Insights into solar wind flow speeds from the coronal radio occultation experiment: Findings from the Indian Mars Orbiter Mission*, **The Astrophysical Journal**, 2025, 10.3847/1538-4357/adb627.
7. Carter, A. L., May, E. M., Espinoza, N., Welbanks, L., et al. (incl. **Aggarwal, K.**), *A benchmark JWST near-infrared spectrum for the exoplanet WASP-39 b*, **Nature Astronomy**, 2024, 10.1038/s41550-024-02292-x.
8. Bell, T. J., Crouzet, N., Cubillos, P., Kreidberg, L., Piette, A. A. A., et al. (incl. **Aggarwal, K.**), *Nightside clouds and disequilibrium chemistry on the hot Jupiter WASP-43b*, **Nature Astronomy**, 2024, 10.1038/s41550-024-02230-x.
9. Powell, D., Feinstein, A., Lee, E., Zhang, M., Tsai, S., Taylor, J., et al. (incl. **Aggarwal, K.**), *Sulphur dioxide in the mid-infrared transmission spectrum of WASP-39b*, **Nature**, 2024, 10.1038/s41586-024-07040-9.
0. Tsai, S., Lee, E., Powell, D., Gao, P., Zhang, X., Moses, J., et al. (incl. **Aggarwal, K.**), *Photochemically produced SO_2 in the atmosphere of WASP-39b*, **Nature**, 2023, 10.1038/s41586-023-05902-2.
11. Grant, D., Lothringer, J. D., Wakeford, H. R., et al. (incl. **Aggarwal, K.**), *Detection of carbon monoxide's 4.6 micron fundamental band structure in WASP-39b's atmosphere with JWST NIRSpec G395H*, **The Astrophysical Journal Letters**, 2023, 10.3847/2041-8213/acd544.

12. Coulombe, L.-P., Benneke, B., Challener, R., Piette, A., Wiser, L., Mansfield, M., et al. (incl. **Aggarwal, K.**), *A broadband thermal emission spectrum of the ultra-hot Jupiter WASP-18b*, **Nature**, 2023, 10.1038/s41586-023-06230-1.

13. JWST Transiting Exoplanet Community Early Release Science Team: Ahrer, E., Alderson, L., Batalha, N., Bean, J., et al. (incl. **Aggarwal, K.**), *Identification of carbon dioxide in an exoplanet atmosphere*, **Nature**, 2023, 10.1038/s41586-022-05269-w.

14. Rustamkulov, Z., Sing, D., Mukherjee, S., May, E., Kirk, J., Schlawin, E., et al. (incl. **Aggarwal, K.**), *Early Release Science of the exoplanet WASP-39b with JWST NIRSpec PRISM*, **Nature**, 2023, 10.1038/s41586-022-05677-y.

15. Alderson, L., Wakeford, H., Alam, M., Batalha, N., Lothringer, J., Redai, J., et al. (incl. **Aggarwal, K.**), *Early Release Science of the exoplanet WASP-39b with JWST NIRSpec G395H*, **Nature**, 2023, 10.1038/s41586-022-05591-3.

16. Seaton, D., Caspi, A., Casini, R., Downs, C., Gibson, S., Gilbert, H., et al. (incl. **Aggarwal, K.**), *Improving multi-dimensional data formats, access, and assimilation tools for the twenty-first century*, **Bulletin of the AAS 55(3), Heliophysics 2024 Decadal Survey white paper**, 2023, baas.aas.org/pub/2023n3i361.

17. Caspi, A., Seaton, D., Casini, R., Downs, C., Gibson, S., Gilbert, H., et al. (incl. **Aggarwal, K.**), *COMPLETE: a flagship mission for complete understanding of 3D coronal magnetic energy release*, **Bulletin of the AAS 55(3), Heliophysics 2024 Decadal Survey white paper**, 2023, baas.aas.org/pub/2023n3i048.

18. Caspi, A., Seaton, D., Casini, R., Downs, C., Gibson, S., Gilbert, H., et al. (incl. **Aggarwal, K.**), *Magnetic energy powers the corona: how we can understand its 3D storage & release*, **Bulletin of the AAS 55(3), Heliophysics 2024 Decadal Survey white paper**, 2023, baas.aas.org/pub/2023n3i049.

19. Zhang, P., Morgan, J., Oberoi, D., Strauss, D. T., Luo, Y., Kontar, E., Huang, Z., **Aggarwal, K.**, Kumari, A., Datta, A., Morosan, D. E., Botha, G. J. J., *Probing the solar corona and the solar wind using angular broadening observations with the SKA*, **Advancing Astrophysics: Preparing for Science with the SKAO**, 2026, arXiv:2603.25421.

20. Choudhary, R. K., **Aggarwal, K.**, Potdar, A., *Long-term GNSS loss-of-lock across the Indian equatorial ionization anomaly*, **GPS Solutions**, 2026 (Under review).

21. Choudhary, R. K., Potdar, A., **Aggarwal, K.**, *High-spatial and temporal 3D imaging of the ionosphere: a GAGAN-based virtual receiver approach*, **Journal of Geophysical Research: Space Weather**, 2026 (Under review).

22. Deep, A., Datta, A., **Aggarwal, K.**, Brawar, B., *Earth's magnetopause response during the Mother's Day storm based on multi-satellite data*, **Solar Physics**, 2026 (Under review).

23. **Aggarwal, K.**, Datta, A., Choudhary, R. K., *Studying the 'Smiley Sun' using the Akatsuki radio occultation experiment*, **Proceedings of the 7th URSI Asia-Pacific Radio Science Conference (AP-RASC 2025)**, 2025, 10.46620/ursiaprasc25/ahct9245.

24. **Aggarwal, K.**, Choudhary, R. K., Datta, A., *Results from radio occultation studies using Indian Mars Orbiter Mission*, **Proceedings of the 6th URSI Regional Conference on Radio Science (RCRS 2024)**, 2024, 10.46620/ursi_rsrc24/1017qon9346.

25. Deep, A., Datta, A., Brawar, B., **Aggarwal, K.**, *Dynamics of Earth's magnetopause during solar storms: insights from in-situ measurements of NASA's MMS mission*, **Proceedings of the 7th URSI Asia-Pacific Radio Science Conference (AP-RASC 2025)**, 2025, 10.46620/ursiaprasc25/qzpe5277.

26. Shekhar, K. S., Tanti, H. A., Datta, A., **Aggarwal, K.**, *Monitoring infrastructure faults with YOLOv5, assisting safety inspectors*, **2023 International Conference on Integration of Computational Intelligent System (ICICIS), IEEE**, 2023, 10.1109/ICICIS56802.2023.10430270.